

Lab Assignment 1

Student Name: Ribhav Devprasad

Roll Number: 261100690002

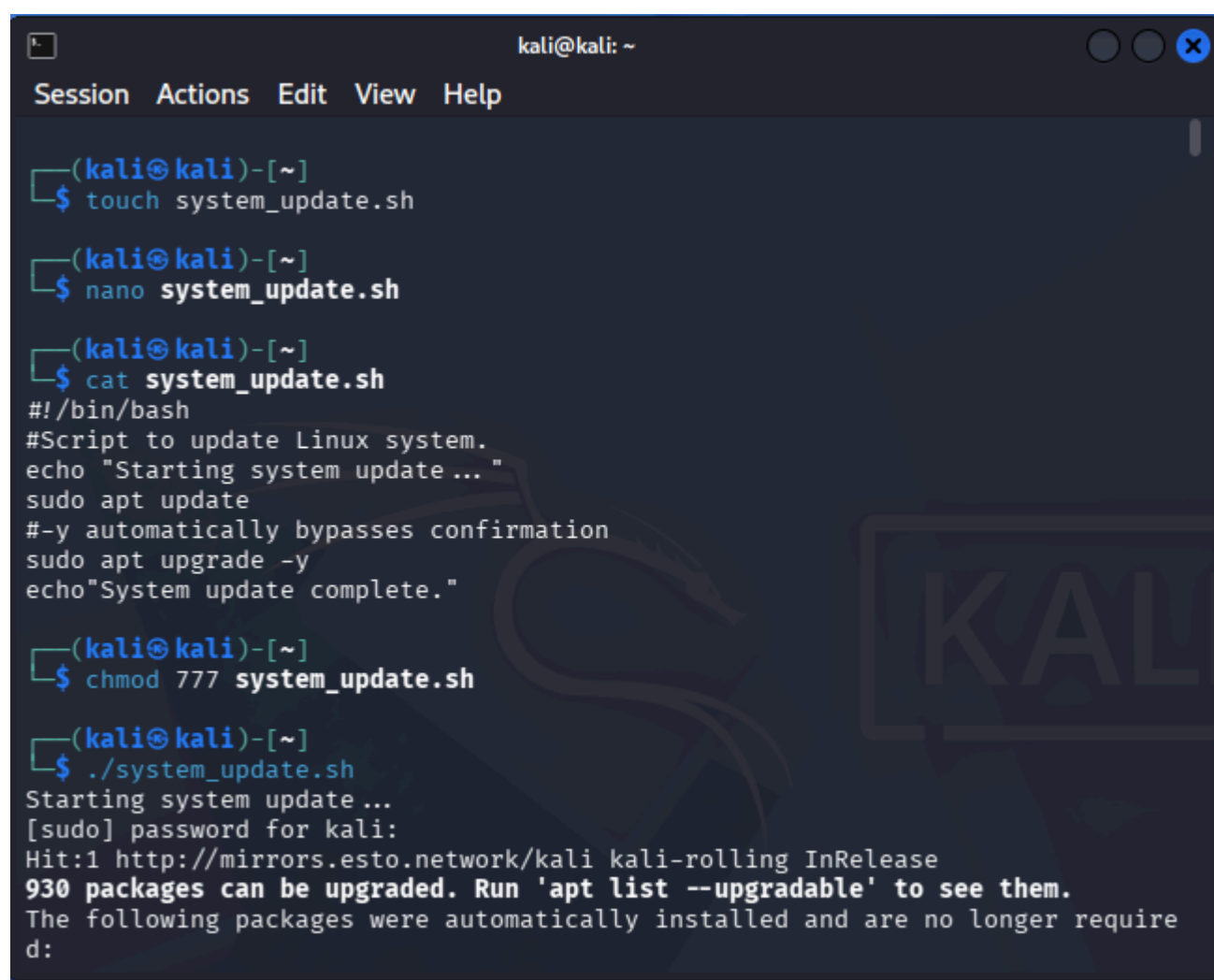
Environment: Kali Linux

1. Write a script to update the system.

Script Code (`system_update.sh`)

```
#!/bin/bash
#Script to update Linux system.
echo "Starting system update ..."
sudo apt update
#-y automatically bypasses confirmation
sudo apt upgrade -y
echo "System update complete."
```

Terminal Output



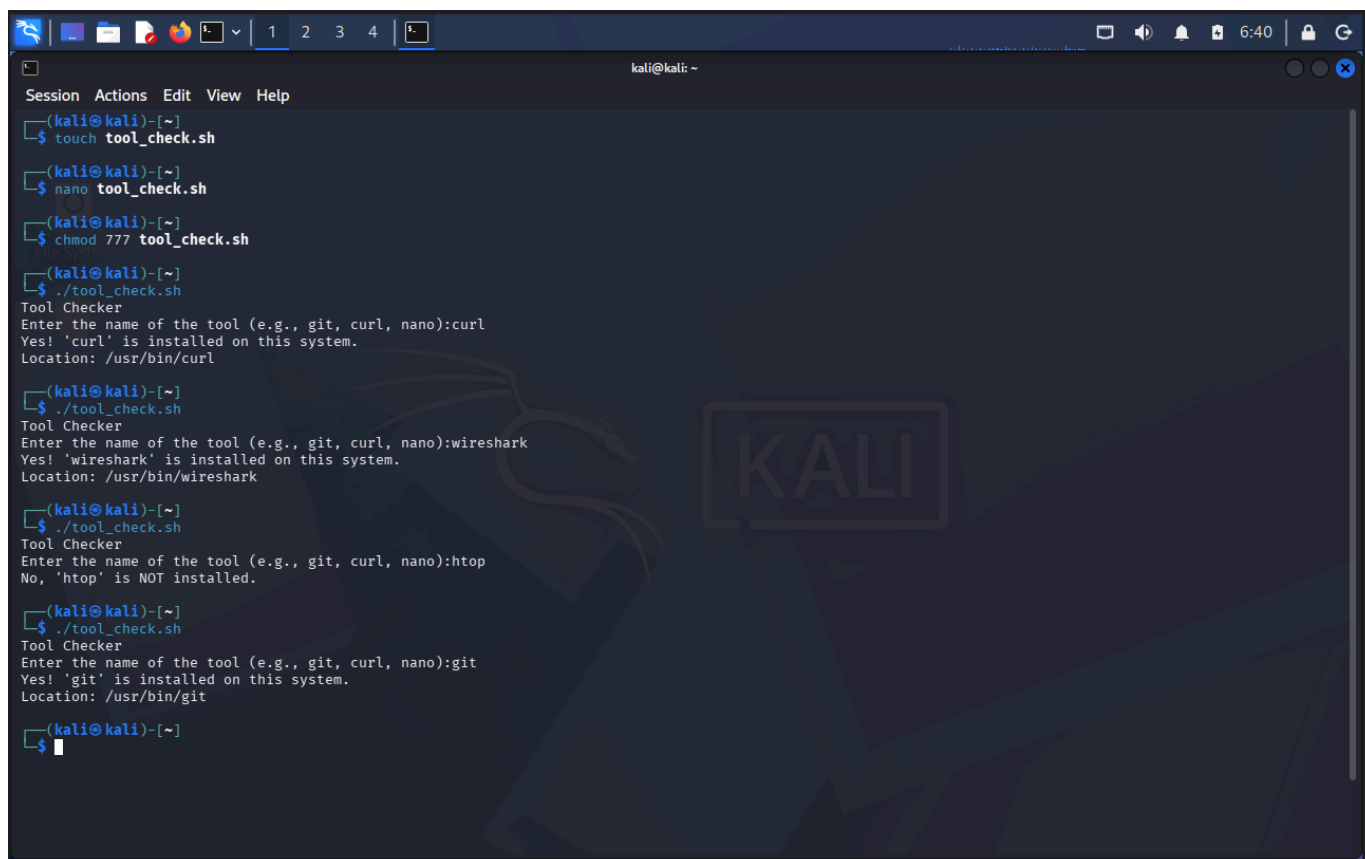
```
kali@kali: ~  
Session Actions Edit View Help  
  
(kali@kali)-[~]  
$ touch system_update.sh  
  
(kali@kali)-[~]  
$ nano system_update.sh  
  
(kali@kali)-[~]  
$ cat system_update.sh  
#!/bin/bash  
#Script to update Linux system.  
echo "Starting system update... "  
sudo apt update  
#-y automatically bypasses confirmation  
sudo apt upgrade -y  
echo "System update complete."  
  
(kali@kali)-[~]  
$ chmod 777 system_update.sh  
  
(kali@kali)-[~]  
$ ./system_update.sh  
Starting system update...  
[sudo] password for kali:  
Hit:1 http://mirrors.esto.network/kali kali-rolling InRelease  
930 packages can be upgraded. Run 'apt list --upgradable' to see them.  
The following packages were automatically installed and are no longer require  
d:
```

2. Write a shell script to get the name of the tool from the user and check whether the tool is installed or not.

Script Code (`tool_check.sh`)

```
#!/bin/bash
#Tool checker script
echo "Tool Checker"
read -p "Enter the name of the tool (e.g., git, curl, nano):" tool
# Check if the tool exists
if which "$tool" &> /dev/null; then
    echo "Yes! '$tool' is installed on this system."
    echo "Location: $(which "$tool")"
else
    echo "No, '$tool' is NOT installed."
fi
```

Terminal Output



```
kali@kali: ~
Session Actions Edit View Help
(kali@kali)-[~]
$ touch tool_check.sh
(kali@kali)-[~]
$ nano tool_check.sh
(kali@kali)-[~]
$ chmod 777 tool_check.sh
(kali@kali)-[~]
$ ./tool_check.sh
Tool Checker
Enter the name of the tool (e.g., git, curl, nano):curl
Yes! 'curl' is installed on this system.
Location: /usr/bin/curl
(kali@kali)-[~]
$ ./tool_check.sh
Tool Checker
Enter the name of the tool (e.g., git, curl, nano):wireshark
Yes! 'wireshark' is installed on this system.
Location: /usr/bin/wireshark
(kali@kali)-[~]
$ ./tool_check.sh
Tool Checker
Enter the name of the tool (e.g., git, curl, nano):htop
No, 'htop' is NOT installed.
(kali@kali)-[~]
$ ./tool_check.sh
Tool Checker
Enter the name of the tool (e.g., git, curl, nano):git
Yes! 'git' is installed on this system.
Location: /usr/bin/git
(kali@kali)-[~]
$
```